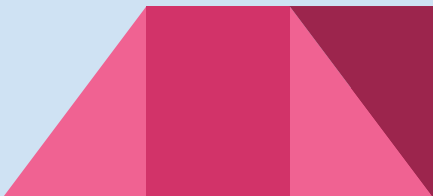




Lecture 8: Sequences & Summations

Section 2.4



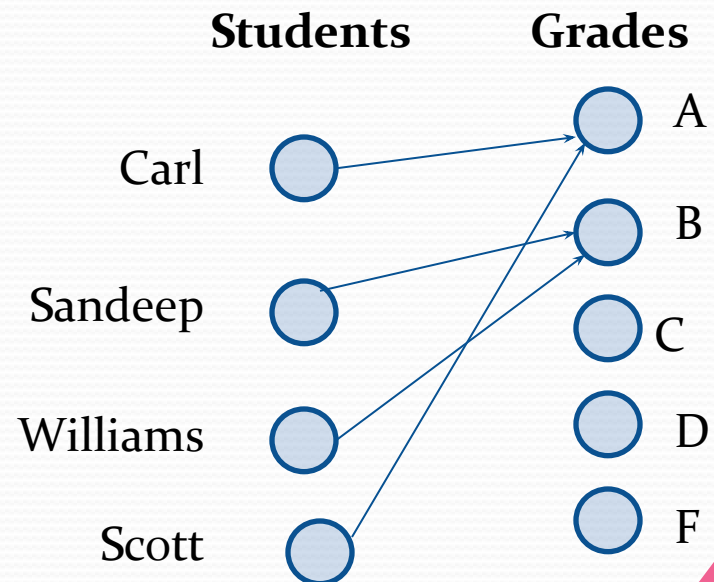
Section Summary

- Sequences.
 - Examples: Geometric Progression, Arithmetic Progression
- Recurrence Relations
 - Example: Fibonacci Sequence
- Summations

Functions

Definition : Let A and B be nonempty sets. A *function* f from A to B , denoted $f: A \rightarrow B$ is an assignment of each element of A to exactly one element of B . We write $f(a) = b$ if b is the unique element of B assigned by the function f to the element a of A .

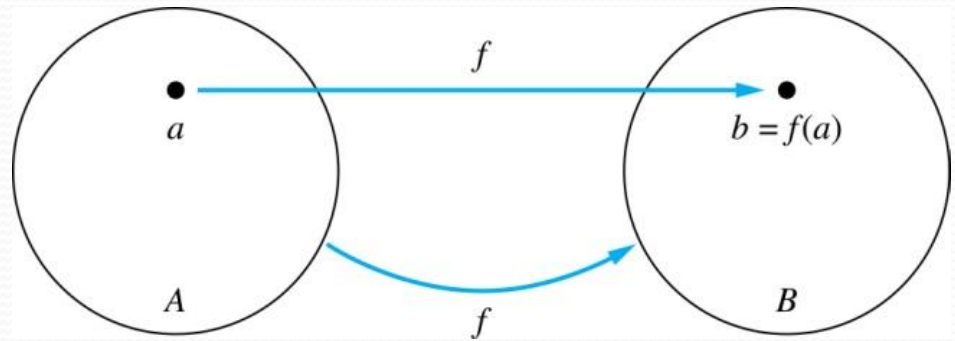
- Functions are sometimes called *mappings* or *transformations*.



Functions

Given a function $f: A \rightarrow B$:

- We say f maps A to B or f is a *mapping* from A to B .
- A is called the **domain** of f .
- B is called the **codomain** of f .
- If $f(a) = b$,
 - then b is called the **image** of a under f .
 - a is called the **preimage** of b .
- The **range** of f is the set of all images of points in A under f . We denote it by $f(A)$.



Sequences

- Sequences are **ordered lists** of elements.
 - 1, 2, 3, 5, 8 (finite)
 - 1, 3, 9, 27, 81, (infinite)

Definition : A *sequence* is a **function** from a subset of the integers (usually either the set $\{0, 1, 2, 3, 4, \dots\}$ or $\{1, 2, 3, 4, \dots\}$) to a set S .

- The notation a_n is used to denote the **image** of the integer n . We can think of a_n as the equivalent of $f(n)$ where f is a function from $\{0, 1, 2, \dots\}$ to S . We call a_n a **term** of the sequence.

Sequences

Example : Consider the sequence $\{a_n\}$ where $a_n = \frac{1}{n}$

$$\{a_n\} = \{a_1, a_2, a_3, \dots\}$$

$$1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4} \dots$$

Geometric Progression

Definition : A *geometric progression* is a sequence of the form:

$$a, ar, ar^2, \dots, ar^n, \dots$$

where the *initial term* a and the *common ratio* r are real numbers.

Examples :

1. Let $a = 1$ and $r = -1$. Then:

$$\{b_n\} = \{b_0, b_1, b_2, b_3, b_4, \dots\} = \{1, -1, 1, -1, 1, \dots\}$$

2. Let $a = 2$ and $r = 5$. Then:

$$\{c_n\} = \{c_0, c_1, c_2, c_3, c_4, \dots\} = \{2, 10, 50, 250, 1250, \dots\}$$

3. Let $a = 6$ and $r = 1/3$. Then:

$$\{d_n\} = \{d_0, d_1, d_2, d_3, d_4, \dots\} = \{6, 2, \frac{2}{3}, \frac{2}{9}, \frac{2}{27}, \dots\}$$

Arithmetic Progression

Definition : A *arithmetic progression* is a sequence of the form:

$$a, a + d, a + 2d, \dots, a + nd, \dots$$

where the *initial term* a and the *common difference* d are real numbers.

Examples :

1. Let $a = -1$ and $d = 4$:

$$\{s_n\} = \{s_0, s_1, s_2, s_3, s_4, \dots\} = \{-1, 3, 7, 11, 15, \dots\}$$

2. Let $a = 7$ and $d = -3$:

$$\{t_n\} = \{t_0, t_1, t_2, t_3, t_4, \dots\} = \{7, 4, 1, -2, -5, \dots\}$$

3. Let $a = 1$ and $d = 2$:

$$\{u_n\} = \{u_0, u_1, u_2, u_3, u_4, \dots\} = \{1, 3, 5, 7, 9, \dots\}$$

Recurrence Relations

Definition: A *recurrence relation* for the sequence $\{a_n\}$ is an equation that expresses a_n in terms of one or more of the **previous terms of the sequence**, namely, $a_0, a_1, \dots, a_{n-1}$, for all integers n with $n \geq n_0$ where n_0 is a nonnegative integer.

- A sequence is called a *solution* of a recurrence relation if its terms satisfy the recurrence relation.
- The *initial conditions* for a sequence specify the terms that precede the first term where the recurrence relation takes effect.

Recurrence Relations

Example 1: Let $\{a_n\}$ be a sequence that satisfies the recurrence relation $a_n = a_{n-1} + 3$ for $n = 1, 2, 3, 4, \dots$ and suppose that $a_0 = 2$. What are a_1 , a_2 and a_3 ?

[Here $a_0 = 2$ is the initial condition.]

Solution : We see from the recurrence relation that

$$a_1 = a_0 + 3 = 2 + 3 = 5$$

$$a_2 = 5 + 3 = 8$$

$$a_3 = 8 + 3 = 11$$

Recurrence Relations

Example 2: Let $\{a_n\}$ be a sequence that satisfies the recurrence relation $a_n = a_{n-1} - a_{n-2}$ for $n = 2, 3, 4, \dots$ and suppose that $a_0 = 3$ and $a_1 = 5$. What are a_2 and a_3 ?

[Here the initial conditions are $a_0 = 3$ and $a_1 = 5$.]

Solution : We see from the recurrence relation that

$$a_2 = a_1 - a_0 = 5 - 3 = 2$$

$$a_3 = a_2 - a_1 = 2 - 5 = -3$$

Fibonacci Sequence

Definition : We define the *Fibonacci sequence*, $f_0, f_1, f_2, \dots$, by:

- Initial Conditions: $f_0 = 0, f_1 = 1$
- Recurrence Relation: $f_n = f_{n-1} + f_{n-2}$

Example : Find f_2, f_3, f_4, f_5 and f_6 .

Answer:

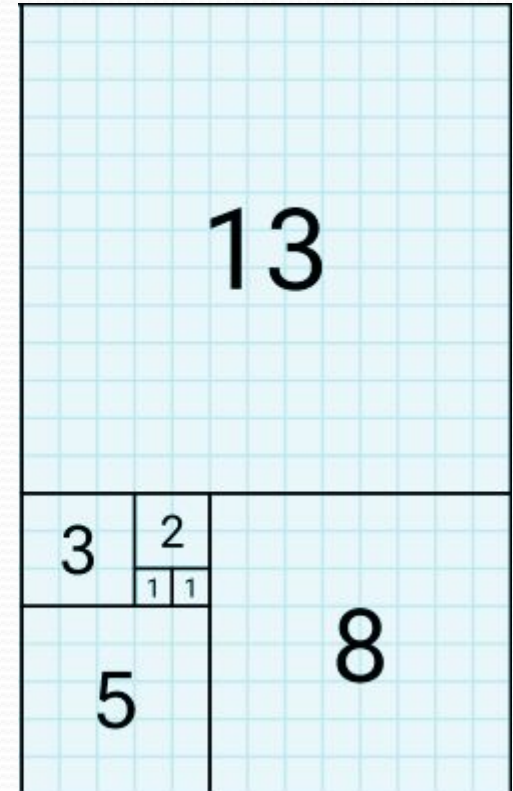
$$f_2 = f_1 + f_0 = 1 + 0 = 1,$$

$$f_3 = f_2 + f_1 = 1 + 1 = 2,$$

$$f_4 = f_3 + f_2 = 2 + 1 = 3,$$

$$f_5 = f_4 + f_3 = 3 + 2 = 5,$$

$$f_6 = f_5 + f_4 = 5 + 3 = 8.$$



Solving Recurrence Relations

- Finding a **formula for the n^{th} term** of the sequence generated by a recurrence relation is called ***solving the recurrence relation*** .
- Such a formula is called a ***closed formula***.
- Various methods for solving recurrence relations will be covered in next lectures (Chapter 8) where recurrence relations will be studied in greater depth.
- For now, we illustrate by example the **method of iteration** in which we need to **guess** the formula. The guess can be proved correct by the method of induction.

Iterative Solution Example

Method 1 : Working upward, forward substitution

Let $\{a_n\}$ be a sequence that satisfies the recurrence relation $a_n = a_{n-1} + 3$ for $n = 2, 3, 4, \dots$ and suppose that $a_1 = 2$.

$$a_2 = 2 + 3$$

$$a_3 = (2 + 3) + 3 = 2 + 3 \cdot 2$$

$$a_4 = (2 + 2 \cdot 3) + 3 = 2 + 3 \cdot 3$$

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$$a_n = a_{n-1} + 3 = (2 + 3 \cdot (n - 2)) + 3 = 2 + 3(n - 1)$$

Iterative Solution Example

Method 2 : Working downward, backward substitution

Let $\{a_n\}$ be a sequence that satisfies the recurrence relation $a_n = a_{n-1} + 3$ for $n = 2, 3, 4, \dots$ and suppose that $a_1 = 2$.

$$\begin{aligned}a_n &= a_{n-1} + 3 \\&= (a_{n-2} + 3) + 3 = a_{n-2} + 3 \cdot 2 \\&= (a_{n-3} + 3) + 3 \cdot 2 = a_{n-3} + 3 \cdot 3 \\&\quad \cdot \\&\quad \cdot \\&\quad \cdot \\&= a_2 + 3(n-2) = (a_1 + 3) + 3(n-2) = 2 + 3(n-1)\end{aligned}$$

Real World Application

Example : Suppose that a person deposits \$10,000.00 in a savings account at a bank yielding 11% per year with interest compounded annually. How much will be in the account after 30 years?

Solution:

Let P_n denote the amount in the account after n years. Then, P_n satisfies the following recurrence relation:

$$P_n = P_{n-1} + 0.11P_{n-1} = (1.11) P_{n-1}$$

with the initial condition $P_0 = 10,000$.

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Real World Application

$$P_n = P_{n-1} + 0.11P_{n-1} = (1.11) P_{n-1}$$

with the initial condition $P_0 = 10,000$

Solving the recurrence with Forward Substitution:

$$P_1 = (1.11)P_0$$

$$P_2 = (1.11)P_1 = (1.11)^2 P_0$$

$$P_3 = (1.11)P_2 = (1.11)^3 P_0$$

:

$$P_n = (1.11)P_{n-1} = (1.11)^n P_0 = (1.11)^n 10,000$$

Since, $P_n = (1.11)^n 10,000$, $P_{30} = (1.11)^{30} 10,000 = \$228,992.97$.

Identifying Integer Sequences

- Given a few terms of a sequence, try to identify the sequence. Conjecture a formula, recurrence relation, or some other rule.
- Some questions to ask?
 - Are there **repeated terms** of the same value?
 - Can you obtain a term from the previous term by **adding** an amount or **multiplying** by an amount?
 - Can you obtain a term by **combining the previous terms** in some way?
 - Are there cycles among the terms?
 - Do the terms match those of a well known sequence?

Identifying Integer Sequences

Find formulae for the sequences with the following first five terms:

Example 1 : 1, $1/2$, $1/4$, $1/8$, $1/16$, ...

Solution: Note that the denominators are powers of 2.

The sequence with $a_n = 1/2^n$ is a possible match.

This is a **geometric** progression with $a = 1$ and $r = 1/2$.

Example 2 : 1, 3, 5, 7, 9

Solution: Note that each term is obtained by adding 2 to the previous term. A possible formula is $a_n = 2n + 1$.

This is an **arithmetic** progression with $a = 1$ and $d = 2$.

Example 3 : 1, -1, 1, -1, 1

Solution: The terms alternate between 1 and -1.

A possible sequence is $a_n = (-1)^n$.

This is a **geometric** progression with $a = 1$ and $r = -1$.

Useful Sequences

TABLE 1 Some Useful Sequences.

<i>n</i> th Term	First 10 Terms
n^2	1, 4, 9, 16, 25, 36, 49, 64, 81, 100, ...
n^3	1, 8, 27, 64, 125, 216, 343, 512, 729, 1000, ...
n^4	1, 16, 81, 256, 625, 1296, 2401, 4096, 6561, 10000, ...
2^n	2, 4, 8, 16, 32, 64, 128, 256, 512, 1024, ...
3^n	3, 9, 27, 81, 243, 729, 2187, 6561, 19683, 59049, ...
$n!$	1, 2, 6, 24, 120, 720, 5040, 40320, 362880, 3628800, ...
f_n	1, 1, 2, 3, 5, 8, 13, 21, 34, 55, 89, ...

Guessing Sequences

Example : Conjecture a simple formula for a_n if the first 10 terms of the sequence $\{a_n\}$ are
1, 7, 25, 79, 241, 727, 2185, 6559, 19681, 59047.

Solution : Note the ratio of each term to the previous approximates 3. So now compare with the sequence 3^n .
 $\{3^n\} = 1, 3, 9, 27, 81, 243, 729, 2187, 6561, 19683, 59049, \dots$
We notice that the n th term is 2 less than the corresponding power of 3. So a good conjecture is that $a_n = 3^n - 2$.

Summations

- Sum of the terms $a_m, a_{m+1}, \dots, a_n$ from the sequence $\{a_n\}$
- The notations:

$$\sum_{j=m}^n a_j \quad \text{or,} \quad \sum_{j=m}^n a_j \quad \text{or,} \quad \sum_{m \leq j \leq n} a_j$$

represents

$$a_m + a_{m+1} + \dots + a_n$$

- The variable j is called the *index of summation*.
It runs through all the integers starting with its *lower limit*, m and ending with its *upper limit*, n .

Summations

- More generally for a set S :

$$\sum_{j \in S} a_j$$

- Examples :**

$$r^0 + r^1 + r^2 + r^3 + \dots + r^n = \sum_0^n r^j$$

$$1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \dots = \sum_1^{\infty} \frac{1}{i}$$

$$\text{If } S = \{2, 5, 7, 10\} \text{ then } \sum_{j \in S} a_j = a_2 + a_5 + a_7 + a_{10}$$

Summations Examples

1. What is the value of $\sum_{j=1}^5 j^2$?

Solution:

$$\begin{aligned}\sum_{j=1}^5 j^2 &= 1^2 + 2^2 + 3^2 + 4^2 + 5^2 \\ &= 1 + 4 + 9 + 16 + 25 \\ &= 55.\end{aligned}$$

1. What is the value of $\sum_{k=4}^8 (-1)^k$?

Solution:

$$\begin{aligned}\sum_{k=4}^8 (-1)^k &= (-1)^4 + (-1)^5 + (-1)^6 + (-1)^7 + (-1)^8 \\ &= 1 + (-1) + 1 + (-1) + 1 \\ &= 1.\end{aligned}$$

Summation of Series Formula

Sums of terms of geometric progressions

$$\sum_{j=0}^n ar^j = \begin{cases} \frac{ar^{n+1} - a}{r - 1} & r \neq 1 \\ (n + 1)a & r = 1 \end{cases}$$

Proof: Let $S_n = \sum_{j=0}^n ar^j$

To compute S_n , first multiply both sides of the equality by r and then manipulate the resulting sum:

$$rS_n = r \sum_{j=0}^n ar^j = \sum_{j=0}^n ar^{j+1}$$

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Summation of Series Formula

$$= \sum_{j=0}^n ar^{j+1} \quad \text{From previous slide.}$$

$$= \sum_{k=1}^{n+1} ar^k \quad \text{Shifting the index of summation with } k = j + 1.$$

$$= \left(\sum_{k=0}^n ar^k \right) + (ar^{n+1} - a) \quad \text{Removing } k = n + 1 \text{ term and adding } k = 0 \text{ term.}$$

$$= S_n + (ar^{n+1} - a) \quad \text{Substituting } S \text{ for summation formula}$$

$$\therefore rS_n = S_n + (ar^{n+1} - a)$$

$$S_n = \frac{ar^{n+1} - a}{r - 1} \quad \text{if } r \neq 1 \quad S_n = \sum_{j=0}^n ar^j = \sum_{j=0}^n a = (n + 1)a \quad \text{if } r = 1$$

Some Useful Summation Formulae

TABLE 2 Some Useful Summation Formulae.

Sum	Closed Form
$\sum_{k=0}^n ar^k \ (r \neq 0)$	$\frac{ar^{n+1} - a}{r - 1}, r \neq 1$
$\sum_{k=1}^n k$	$\frac{n(n+1)}{2}$
$\sum_{k=1}^n k^2$	$\frac{n(n+1)(2n+1)}{6}$
$\sum_{k=1}^n k^3$	$\frac{n^2(n+1)^2}{4}$
$\sum_{k=0}^{\infty} x^k, x < 1$	$\frac{1}{1-x}$
$\sum_{k=1}^{\infty} kx^{k-1}, x < 1$	$\frac{1}{(1-x)^2}$

for Arithmetic progression,
the sum of first n terms:

$$\sum_{i=0}^{n-1} (a + id) = \frac{n}{2} \{2a + (n-1)d\}$$

We can
prove these
by
induction,
also.

Proof requires
calculus